

AWS Cloudformation

What is AWS CloudFormation?

- Service for modeling and setting up AWS resources
- Uses templates written in YAML or JSON
- Enables Infrastructure as Code (IaC)
- Automates provisioning and management

Key Benefits

- **Automation:** No manual setup of resources
- **Consistency:** Same infrastructure across environments
- **Version Control:** Templates can be stored in Git
- **Cost Management:** Track and replicate infrastructure more easily

How CloudFormation Works

- Define resources in a template
- Upload to CloudFormation
- CloudFormation provisions resources in the right order
- Supports rollback and update mechanisms

Template Structure

✓ Sections:

- ✓ AWSTemplateFormatVersion
 - ✓ Description
 - ✓ Parameters
 - ✓ Resources (core)
 - ✓ Outputs
- ✓ Written in YAML (preferred) or JSON

Example Template

Resources:

MyS3Bucket:

Type: AWS::S3::Bucket

Properties:

BucketName: my-example-bucket

Parameters and Outputs

- **Parameters:** Input values for dynamic templates
- **Outputs:** Export important information (e.g., ARN, URL)
- Improves reusability and modularity

Stack Management

- **Stacks:** Deployed instances of templates
- **Stack Sets:** Manage stacks across multiple accounts/regions
- Lifecycle: Create → Update → Delete
- Rollback support on failure

Advanced Features

- **Nested Stacks:** Organize complex templates
- **Drift Detection:** Check for manual changes
- **Change Sets:** Preview changes before deployment
- **Integration with CI/CD pipelines**

Summary & Resources

- CloudFormation simplifies and automates infrastructure management
- Encourages IaC best practices
- **Resources:**
 - AWS Docs: <https://docs.aws.amazon.com/cloudformation>
 - AWS Free Tier for hands-on practice